

8.1 Ripping Tracks

Before you can play any music on your fancy player, you've got to have some digital music to transfer. It's time to rip - geek-speak for copying a track (say, from a CD) to your hard drive. Some players let you rip without a PC, but you'll often get better quality if you rip tracks to your PC first and transfer them to the player later.

Digital audio file formats such as MP3 (MPEG 1, Layer 3) and WMA (Windows Media Audio) compress bulky WAV versions of audio tracks into much smaller files. You'll find quite a few alternatives when it comes to ripping: MP3, WMA, RealAudio, LiquidAudio, Ogg Vorbis, and more. So how do you pick?

Let your player be your guide. Which formats does it play? All players can read MP3 files, and most can now play WMA files, so one of those two is probably your best bet. WMA files sound better at lower bit rates (a yardstick of quality and file size), but the MP3 format is far more popular and isn't subject to digital rights management that can prevent you from copying multiple versions of a file.

When ripping a file you'll select a format, then you'll need to consider the bit rate you want. In general, the higher the bit rate, the better the sound and the larger the file. Unless you're a gold-eared audiophile, an MP3 file encoded at 128 or 160 kilobits per second should be fine for a portable player; a WMA file at 96 or 128 kbps would be the equivalent and take up less memory - a major consideration on flash memory players.

8.2 Burning Data Onto CD

If you don't back up your data, you should. It's easy and can save you a lot of pain later if your PC eats a file or if you accidentally drop your hard drive out of the window. Because CD-Rs and CD-RWs hold only a maximum 700 MB of data, I'd recommend putting your documents on one CD, and other kinds of things, say your e-mail archives, on another. You shouldn't need to back up your various applications, because you should have the original